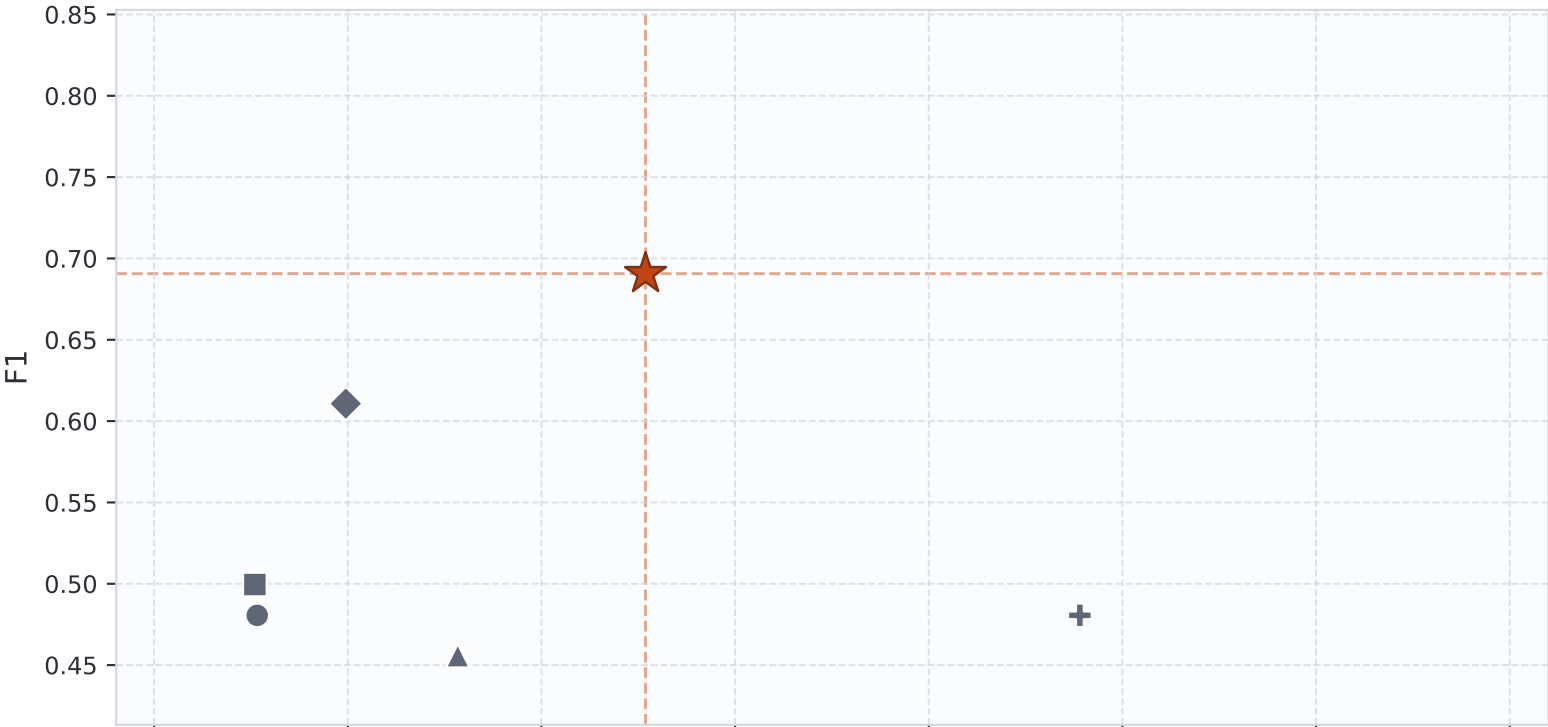


RAG Family vs GRAG: Time vs F1

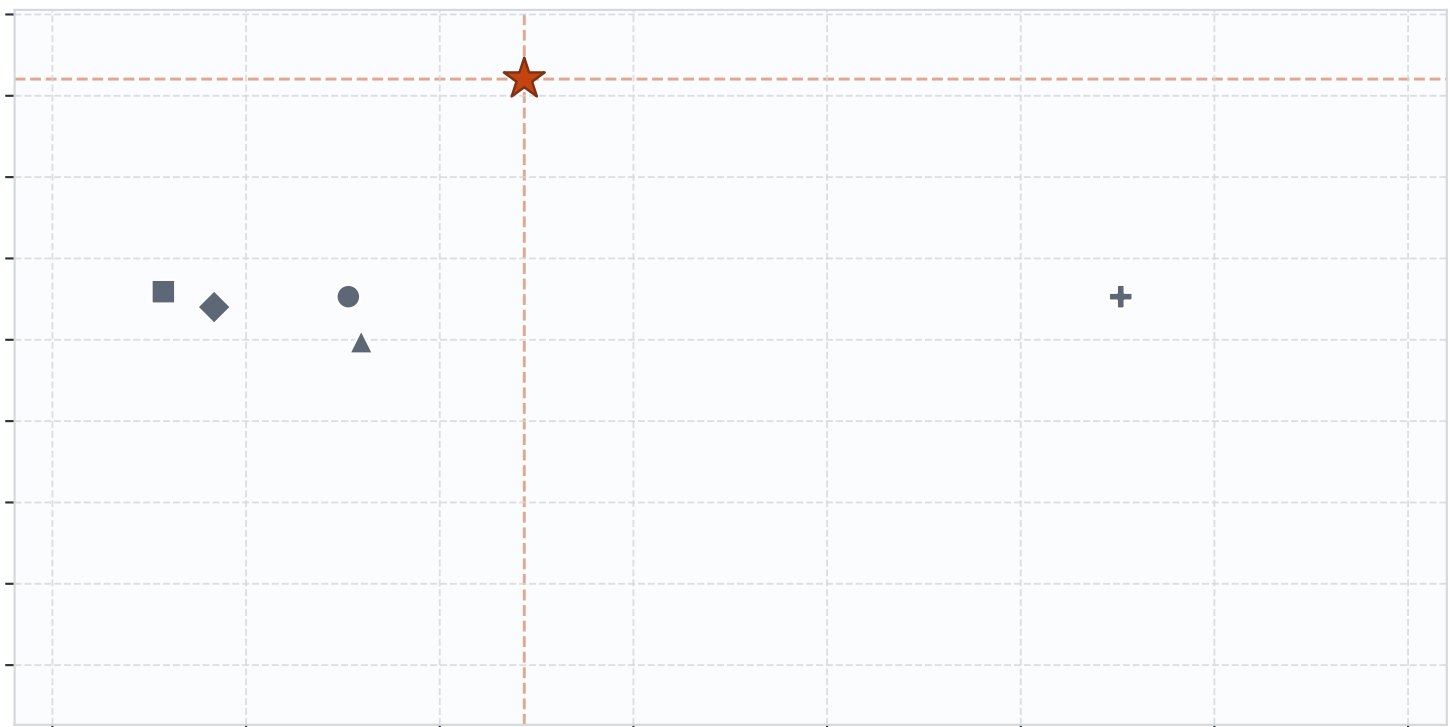
Higher and farther left is better. Compares end-to-end latency with retrieval F1.

- ★ GRAG
- Naive
- Flat Hybrid
- ▲ Hierarchical
- ◆ Parent-Child
- + Query Rewrite
- - - GRAG reference lines

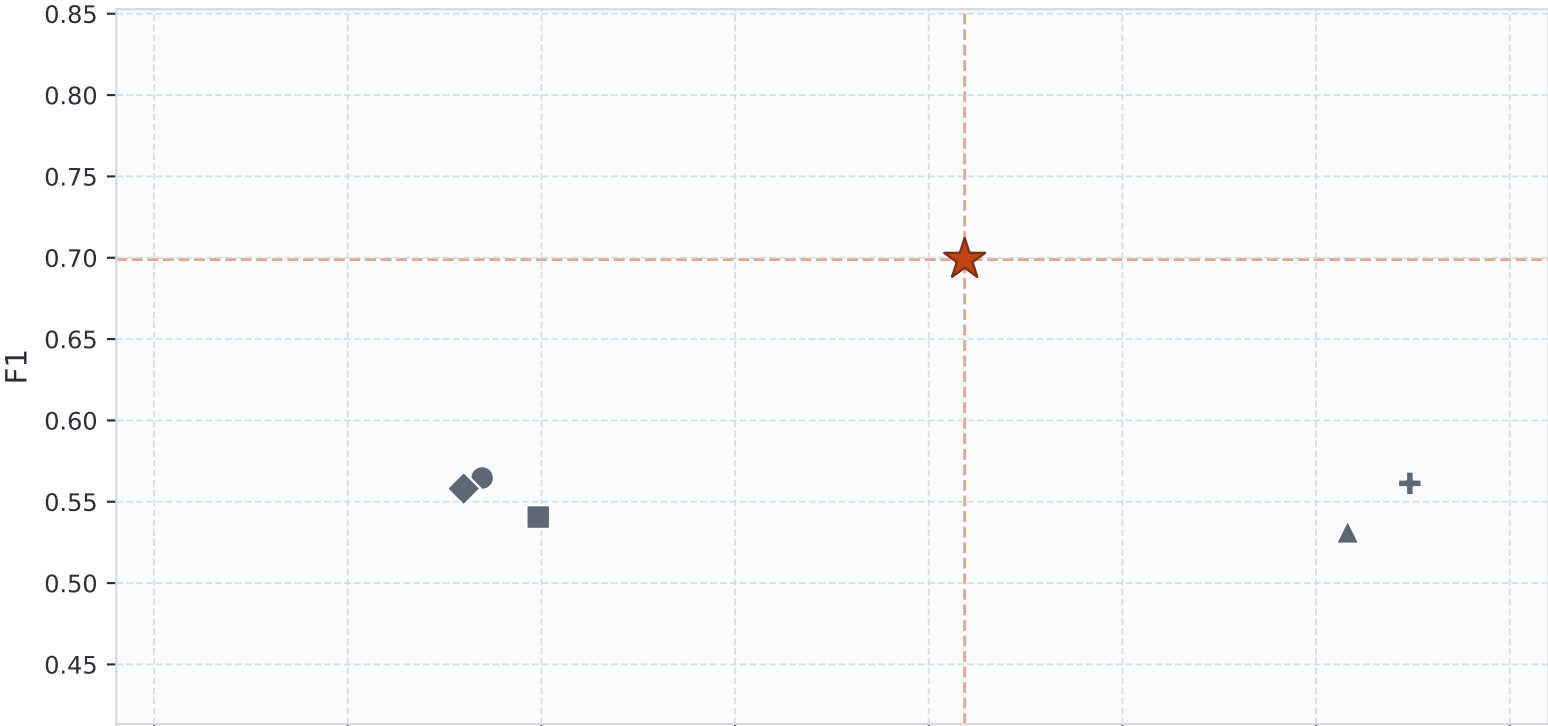
BGE-M3 (ONNX)
Gemma-4-E2B



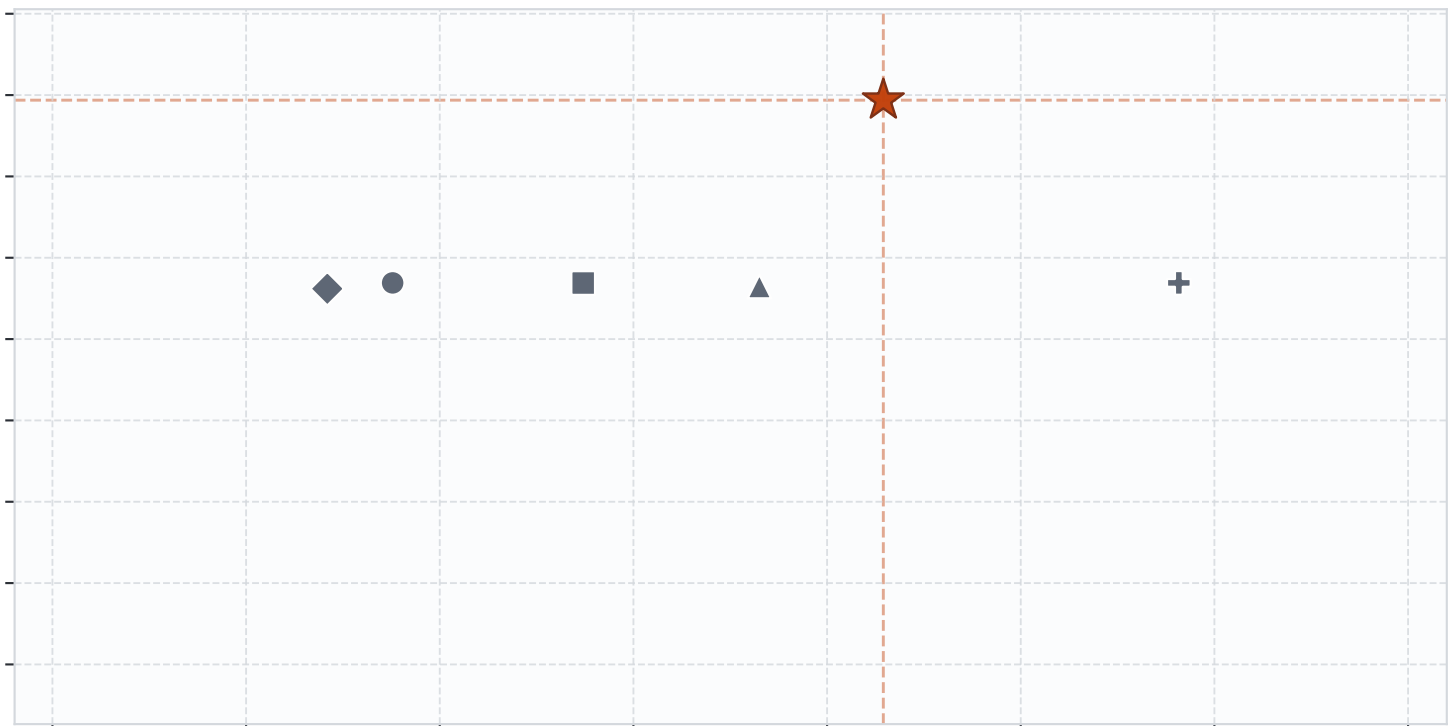
BGE-M3 (ONNX)
Qwen3.5-2B



Qwen3-Embedding 0.6B (ONNX)
Gemma-4-E2B



Qwen3-Embedding 0.6B (ONNX)
Qwen3.5-2B



Average End-to-End Time per Question (s)

Average End-to-End Time per Question (s)